

DS HOMEWORK 3

1) Write a C program to store \$n\$ elements in an array using dynamic memory allocation and find largest number.

CODE:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int *arr,n,i;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    arr =(int *)malloc(n*sizeof(int));
    printf("Enter the array elements:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int high=arr[0];
    for(i=1;i<n;i++)
    {
        if(arr[i]>high)
            high=arr[i];
    }
    printf("Largest Element:%d",high);
    return 0;
}
```

OUTPUT:

Enter the number of elements:3

Enter the array elements:

1

2

3

Largest Element:3

2) Print an array in reverse order using pointers.

CODE:

```
#include<stdio.h>
int main()
{
    int n,i;
```

```

printf("Enter the number of elements:");
scanf("%d",&n);
int arr[n];
printf("Enter the array Elements:\n");
for(i=0;i<n;i++)
{
    scanf("%d",&arr[i]);
}
printf("Array in reverse order:");
int *ptr=&arr[n-1];
for(i=0;i<n;i++)
{
    printf("%d  ",*ptr);
    ptr--;
}
return 0;
}

```

OUTPUT:

Enter the number of elements:3

Enter the array Elements:

1

2

3

Array in reverse order:3 2 1

3) Write a C program to store information of students using structure. Sort them by Name.

CODE:

```

#include<stdio.h>
#include<string.h>
struct student
{
    int rollno;
    char name[25];
};
int main()
{
    int n;
    printf("Enter the number of students:");
    scanf("%d",&n);
    struct student S[n];
}

```

```

for(int i=0;i<n;i++)
{
    printf("Student %d\n",i+1);
    printf("Roll no:");
    scanf("%d",&S[i].rollno);
    printf("Name:");
    scanf("%s",S[i].name);
}
struct student temp;
for(int i=0;i<n-1;i++)
{
    for(int j=i+1;j<n;j++)
    {
        if(strcmp(S[i].name,S[j].name)>0)
        {
            temp=S[i];
            S[i]=S[j];
            S[j]=temp;
        }
    }
}
for(int i=0;i<n;i++)
{
    printf("Sorted:\n");
    printf("Roll no:%d\n",S[i].rollno);
    printf("Name:%s\n",S[i].name);
}
}

```

OUTPUT:

Enter the number of students:2

Student 1

Roll no:52

Name:yoga

Student 2

Roll no:92

Name:visalini

Sorted:

Roll no:92

Name:visalini

Sorted:

Roll no:52

Name:yoga

4) Write a C program to add \$n\$ distances given in inch-feet system using structures.

CODE:

```
#include<stdio.h>
struct distance
{
    int inch;
    int feet;
};
int main()
{
    int n;
    printf("Enter the number of distaces:");
    scanf("%d",&n);
    struct distance D[n];
    int sumf=0,sumi=0;
    for(int i=0;i<n;i++)
    {
        printf("Distance %d\n",i+1);
        printf("Feet:");
        scanf("%d",&D[i].feet);
        printf("Inch:");
        scanf("%d",&D[i].inch);
        sumf+=D[i].feet;
        sumi+=D[i].inch;
    }
    sumf+=sumi/12;
    sumi=sumi%12;
    printf("\nTotal distance:%d feet %d inches",sumf,sumi);
}
```

OUTPUT: Enter the number of distaces:2

Distance 1

Feet:12

Inch:34

Distance 2

Feet:54

Inch:76

Total distance:75 feet 2 inches

5) Write a C program to add two complex numbers by passing structure to a function.

CODE:

```
#include<stdio.h>
struct Complex
{
    int real;
    int img;
};
struct Complex add(struct Complex c1,struct Complex c2)
{
    struct Complex s;
    s.real=c1.real+c2.real;
    s.img=c1.img+c2.img;
    return s;
}
int main()
{
    struct Complex c1,c2,s;
    printf("Enter the real parts:");
    scanf("%d %d",&c1.real,&c2.real);
    printf("Enter the imaginary parts:");
    scanf("%d %d",&c1.img,&c2.img);
    s=add(c1,c2);
    printf("Sum of real part %d:",s.real);
    printf("\nSum of imaginary part %d:",s.img);
    return 0;
}
```

OUTPUT:

Enter the number of distaces:2

Distance 1

Feet:12

Inch:34

Distance 2

Feet:54

6) Write a C program to maintain inventory management system using structure.

CODE:

```
#include<stdio.h>
struct inventory
{
    int id;
    char name[25];
    int qty;
    float price;
};
void add(struct inventory i[],int *n)
{
    printf("\nEnter ID:");
    scanf("%d",&i[*n].id);
    printf("\nEnter name:");
    scanf("%s",i[*n].name);
    printf("\nEnter Quantity:");
    scanf("%d",&i[*n].qty);
    printf("\nEnter price:");
    scanf("%f",&i[*n].price);
    (*n)++;
    printf("\nProduct Added");
}
void display(struct inventory i[],int n)
{
    int j;
    if(n==0)
    {
        printf("\nNo products available.\n");
        return;
    }
    printf("\nID\tName\tQuantity\tPrice\n");
    for(j = 0; j < n; j++)
    {
        printf("%d\t%s\t%d\t\t%.2f\n",
            i[j].id, i[j].name,
            i[j].qty, i[j].price);
    }
}
void search(struct inventory i[],int n)
{
    int id,j,found = 0;
    printf("\nEnter Product ID to search: ");
```

```

scanf("%d", &id);
for(j= 0;j< n;j++)
{
    if(i[j].id == id)
    {
        printf("\nProduct Found!\n");
        printf("ID      : %d\n", i[j].id);
        printf("Name     : %s\n", i[j].name);
        printf("Quantity : %d\n", i[j].qty);
        printf("Price    : %.2f\n", i[j].price);
        found = 1;
        break;
    }
}
}
int main()
{
    int n=0,choice;
    struct inventory i[100];
    printf("Inventory Management system\n1.Add product\n2.display
product\n3.Search product\n4.Exit\n");
    do
    {
        printf("\nEnter your choice:");
        scanf("%d",&choice);
        switch(choice)
        {
            case 1:
                add(i,&n);
                break;
            case 2:
                display(i,n);
                break;
            case 3:
                search(i,n);
                break;
            default:
                break;
        }
    }while(choice!=4);
    return 0;
}

```

OUTPUT:

Inventory Management system

- 1.Add product
- 2.display product
- 3.Search product
- 4.Exit

Enter your choice:2

No products available.

Enter your choice:1

Enter ID:23

Enter name:soap

Enter Quantity:5

Enter price:30

Product Added

Enter your choice:4

7) Write a C program for decimal to binary, octal and hexadecimal conversion.

CODE:

```
#include<stdio.h>
void binary(int n)
{
    int i=0,a[32];
    if(n==0)
    {
        printf("0");
        return;
    }
    while(n>0)
    {
        a[i]=n%2;
        n=n/2;
        i++;
    }
    printf("Binary:");
    for(i=i-1;i>=0;i--)
    {
        printf("%d",a[i]);
    }
}
```



```

void octal(int n)
{
    int i=0,a[32];
    if(n==0)
    {
        printf("0");
        return;
    }
    while(n>0)
    {
        a[i]=n%8;
        n=n/8;
        i++;
    }
    printf("\nOctal:");
    for(i=i-1;i>=0;i--)
    {
        printf("%d",a[i]);
    }
}

void hex(int n)
{
    int i=0,a[32],r;
    if(n==0)
    {
        printf("0");
        return;
    }
    while(n>0)
    {
        r=n%16;
        if(r<10)
            a[i]=r+'0';
        else
            a[i]=r-10+'A';
        n=n/16;
        i++;
    }
    printf("\nHexadecimal:");
    for(i=i-1;i>=0;i--)
    {
        printf("%c",a[i]);
    }
}

int main()
{
    int n;

```

```

printf("Enter a Number:");
scanf("%d",&n);
binary(n);
octal(n);
hex(n);
return 0;
}

```

OUTPUT:

Enter a Number:6478

Binary:1100101001110

Octal:14516

Hexadecimal:194E

8) Write a C program to calculate difference between two time periods.

CODE:

```

#include<stdio.h>
struct time
{
    int hr;
    int min;
    int sec;
};
int main()
{
    struct time t1,t2,t3;
    printf("Enter first time preriod:\n");
    scanf("%d %d %d",&t1.hr,&t1.min,&t1.sec);
    printf("Enter second time preriod:\n");
    scanf("%d %d %d",&t2.hr,&t2.min,&t2.sec);
    int t1s=(t1.hr*3600)+(t1.min*60)+t1.sec;
    int t2s=(t2.hr*3600)+(t2.min*60)+t2.sec;
    int diff=t2s-t1s;
    t3.hr=diff/3600;
    diff=diff%3600;
    t3.min=diff/60;
    t3.sec=diff%60;
    printf("Difference=%d Hrs %d mins %d secs.",t3.hr,t3.min,t3.sec);
    return 0;
}

```

OUTPUT:

Enter first time prerioid:

3

45

32

Enter second time prerioid:

6

45

32

Difference=3 Hrs 0 mins 0 secs.

9) Write a C program to implement an employee management system using structures.

CODE:

```
#include <stdio.h>
struct Employee
{
    int id;
    char name[30];
    float salary;
};
void addEmployee(struct Employee e[], int *n)
{
    printf("\nEnter Employee ID: ");
    scanf("%d", &e[*n].id);
    printf("Enter Employee Name: ");
    scanf("%s", e[*n].name);
    printf("Enter Salary: ");
    scanf("%f", &e[*n].salary);
    (*n)++;
    printf("\nEmployee added successfully!\n");
}
void displayEmployees(struct Employee e[], int n)
{
    int i;
    if(n == 0)
    {
        printf("\nNo employees available.\n");
        return;
    }
}
```

```

    printf("\nID\tName\tSalary\n");
    for(i = 0; i < n; i++)
    {
        printf("%d\t%s\t%.2f\n",e[i].id,e[i].name,e[i].salary);
    }
}

void searchEmployee(struct Employee e[], int n)
{
    int id, i, found = 0;
    printf("\nEnter Employee ID to search: ");
    scanf("%d", &id);
    for(i = 0; i < n; i++)
    {
        if(e[i].id == id)
        {
            printf("\nEmployee Found!\n");
            printf("ID      : %d\n", e[i].id);
            printf("Name   : %s\n", e[i].name);
            printf("Salary : %.2f\n", e[i].salary);
            found = 1;
            break;
        }
    }
    if(found == 0)
        printf("\nEmployee not found.\n");
}

int main()
{
    struct Employee e[100];
    int n = 0, choice;
    do
    {
        printf("\nEMPLOYEE MANAGEMENT SYSTEM");
        printf("\n1.Add Employee");
        printf("\n2.Display Employees");
        printf("\n3.Search Employee");
        printf("\n4.Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);
        switch(choice)
        {
            case 1:
                addEmployee(e, &n);
                break;
            case 2:
                displayEmployees(e, n);
                break;

```

```

        case 3:
            searchEmployee(e, n);
            break;
        default:
            break;
    }
} while(choice != 4);
return 0;
}

```

OUTPUT:

EMPLOYEE MANAGEMENT SYSTEM

1. Add Employee
2. Display Employees
3. Search Employee
4. Exit

Enter your choice: 1

Enter Employee ID: 101

Enter Employee Name: Arun

Enter Salary: 25000

Employee added successfully!

Enter your choice: 4

10) Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

CODE:

```

#include <stdio.h>
#include <math.h>
void rectanglePerimeter(int *l, int *b)
{
    printf("Perimeter of rectangle: %d", 2 * (*l) + 2 * (*b));
}
void trianglePerimeter(int *s1, int *s2, int *s3)
{
    printf("\nPerimeter of triangle: %d", *s1 + *s2 + *s3);
}
void rectangleArea(int *l, int *b)

```

```

{
    printf("\nArea of rectangle: %d", (*l) * (*b));
}
void triangleArea(int *s, int *s1, int *s2, int *s3)
{
    printf("\nArea of triangle: %.2f",sqrt((*s) * (*s - *s1) *(*s - *s2) * (*s
- *s3)));
}
int main()
{
    int l, b;
    int s1, s2, s3;
    int s;
    printf("Enter the dimensions of rectangle: ");
    scanf("%d %d", &l, &b);
    printf("Enter the dimensions of triangle: ");
    scanf("%d %d %d", &s1, &s2, &s3);
    s = (s1 + s2 + s3) / 2;
    rectanglePerimeter(&l, &b);
    trianglePerimeter(&s1, &s2, &s3);
    rectangleArea(&l, &b);
    triangleArea(&s, &s1, &s2, &s3);
    return 0;
}

```

OUTPUT:

Enter the dimensions of rectangle: 2

3

Enter the dimensions of triangle: 4

5

6

Perimeter of rectangle: 10

Perimeter of triangle: 15

Area of rectangle: 6

Area of triangle: 6.48
